Instructions for *ACL Proceedings

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Abstract

(SP: I basically just copied Jeff's abstract with a few edits. Probably should be shortened considerably) Soubki and Heinz (2023) examine the performance of the state-merging algorithms RPNI, EDSM, and ALERGIA on an early version of MLRegTest, a benchmark for the learning of regular languages (van der Poel et al., 2024). They found that the neural models generally outperformed the state-merging algorithms on this benchmark. However, the MLRegTest benchmark excludes strings less than length 19 in order to achieve a balance of positive and negative strings in training, development and test sets across all languages in MLRegTest. Soubki and Heinz (2023) report a follow-up experiment which includes shorter strings and show that they dramatically improve the performance of the state-merging algorithms.

This paper makes three contributions. First, it replicates the experiments in Soubki and Heinz (2023) for the neural networks and state-merging algorithms RPNI and EDSM on the final version of MLRegTest. This replication includes a grid search over many parameters belonging to the state-merging algorithms provided by the FlexFringe software (Verwer and Hammerschmidt, 2017, 2022) to find suitable hyperparameters. Second, it conducts a comprehensive set of follow-experiments with shorter strings on *both* the neural and state-merging learning systems to allow for comparison. Third, on the basis of these experiments, we conclude that the neural systems outperform the state-merging systems in all conditions save one. When the training data is small, and the short strings are included, EDSM outperforms the neural networks.

1 Introduction

2 MLRegTest

(SP: Maybe this section should be something more like "evaluation" or "data" given that the OS and PS are in addition to the standard MLRegTest)

3 Learning Systems

3.1 State-Merging Systems

Following Soubki and Heinz (2023), we make use of the state-merging algorithms provided by FlexFringe (Verwer and Hammerschmidt, 2017, 2022). We build on Soubki and Heinz by conducting a limited gridsearch over the following parameters provided by FlexFringe:

- PerformFirst: corresponding to whether the algorithm should perform EDSM or a modified version of RPNI
- Extend (TODO: what does this do)
- ShallowFirst (TODO: what does this do)
- SearchStrategy: searchdeep, searchlocal, searchglobal, searchpartial, none (TODO: discuss)

We also tried to tune the following parameters but it caused segfaulting: ReverseTraces, DepthCheck, SymbolCheck, and SinksOn, SinkCount, and MergeScore. All of these parameters thus remained at their default values.

(TODO: a word about how we gridsearched)

3.2 Neural Systems

Following van der Poel et al. (2024), we test four neural architectures, all consisting of a trainable embedding layer followed by a unidirectional recurrent module (either a **Simple** RNN, a **GRU**, or an **LSTM**) or two consecutive **Transformer** blocks, dense feed-forward layers, dropout, layer normalization, and softmax activation. All neural networks are implemented with TensorFlow and Keras (TODO: cite packages) and use the same hyperparameters as van der Poel et al. (2024).[1]

[1](TODO: add link to the repo)

4 Experiments

5 Results

6 Discussion

7 Conclusions

References

Adil Soubki and Jeffrey Heinz. 2023. Benchmarking state-merging algorithms for learning regular languages. In *Proceedings of 16th edition of the International Conference on Grammatical Inference*, volume 217 of *Proceedings of Machine Learning Research*, pages 181–198. PMLR.

Sam van der Poel, Dakotah Lambert, Kalina Kostyszyn, Tiantian Gao, Rahul Verma, Derek Andersen, Joanne Chau, Emily Peterson, Cody St. Clair, Paul Fodor, Chihiro Shibata, and Jeffrey Heinz. 2024. Mlregtest: A benchmark for the machine learning of regular languages. *Journal of Machine Learning Research*, 25(283):1–45.

Sicco Verwer and Christian Hammerschmidt. 2022. FlexFringe: Modeling software behavior by learning probabilistic automata. *arXiv preprint*.

Sicco Verwer and Christian A. Hammerschmidt. 2017. FlexFringe: A passive automaton learning package. In *2017 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, pages 638–642.

A Example Appendix

This is an appendix.